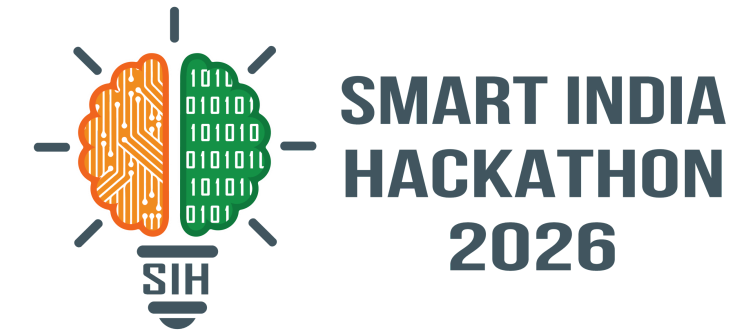


SMART INDIA HACKATHON 2026



- **Problem Statement ID** – SIH26155
- **Problem Statement Title-** AI-Driven Multi-Vendor Network Security Compliance Auditor
- **Theme-** Blockchain & Cybersecurity
- **PS Category-** Software
- **Team ID-** 125474
- **Team Name-** EliteCore



Different Vendors. One Security Language.



THE PROBLEM

Network environments often include devices from vendors such as Cisco, Juniper, Fortinet, Palo Alto, Check Point, and Huawei. Since their configuration commands are different, checking security settings across all these devices takes a lot of manual work and time.



PROPOSED SOLUTION

Our solution uses AI to analyze network configurations from different vendors and convert them into a common security format.

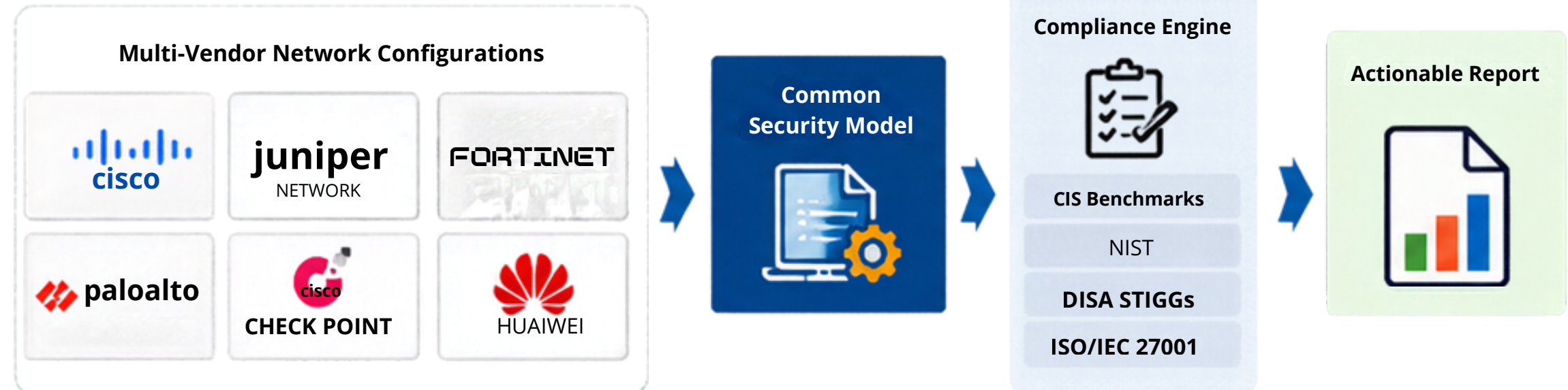


HOW IT WORKS

1. Take configuration files from different network vendors.
2. Find out which device and vendor the configuration belongs to, along with its security settings.
3. Put the different vendor configurations into a common security model.
4. Compare the configuration with the selected security frameworks.
5. Check which controls have failed and give them a severity level.
6. Suggest suitable fixes for each affected device.
7. Prepare one compliance report with all the results.



FROM DIVERSE CONFIGURATIONS TO ONE COMPLIANCE VIEW

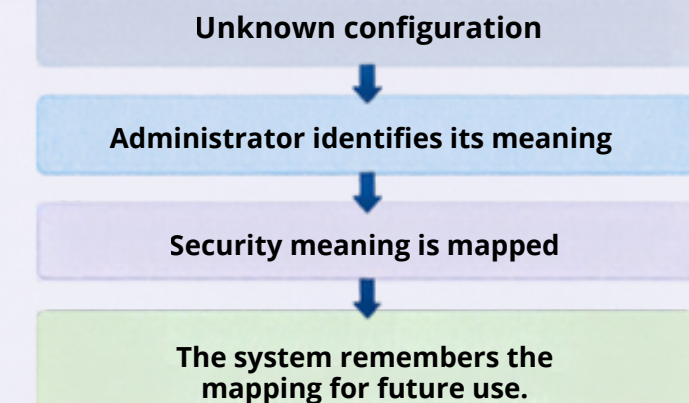


SECURITY FRAMEWORKS



KEY INNOVATION

The system can understand unfamiliar configuration structures instead of depending only on predefined vendor commands.

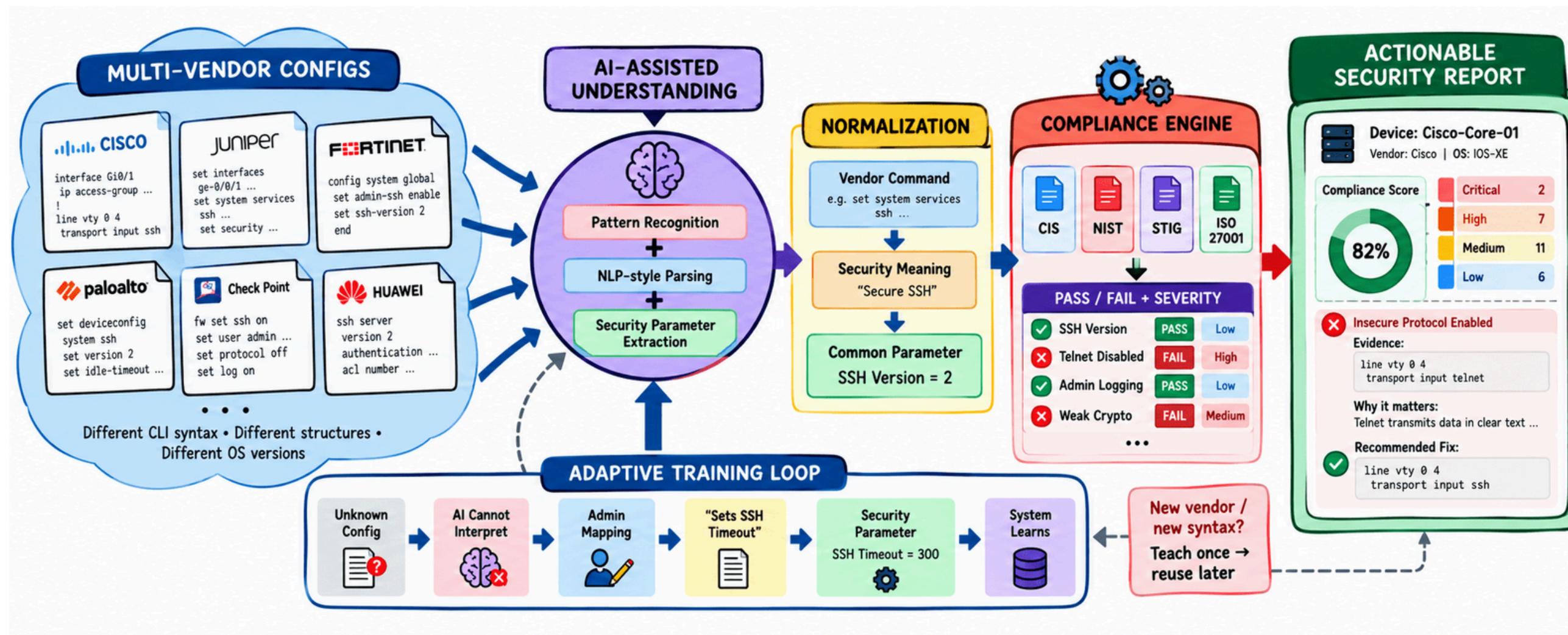


OUR SOLUTION

Most tools check configuration.

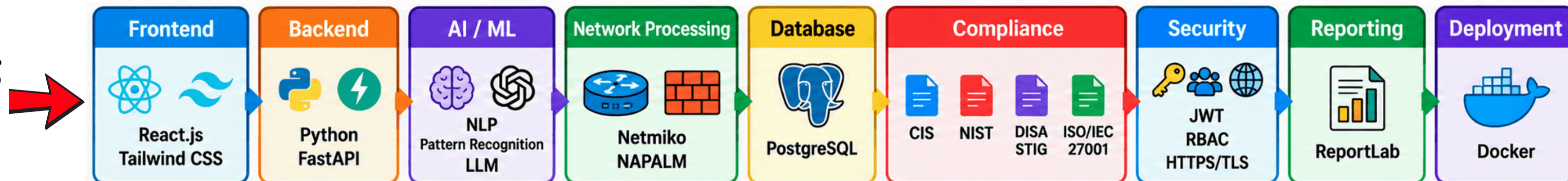
Our approach focuses on Understanding them.

System Architecture of Core Functionality



Our system takes configurations from different network vendors and uses AI-assisted parsing to understand their different command structures. It extracts the security information, converts it into a common security model, and checks it against frameworks such as CIS, NIST, STIG, and ISO 27001. The compliance engine marks each control as PASS or FAIL and assigns a severity level to the findings. These results are then turned into a security report showing the device, compliance score, failed controls, evidence, and recommended fixes for each device. The key feature is the Adaptive Training Loop. If the AI finds an unfamiliar configuration it cannot understand, an administrator can map it to its security meaning. The system learns this mapping and reuses it when similar configurations appear in the future.

Technologies Used



01 FEASIBILITY

The solution is practical to implement using existing software technologies. No specialized hardware is required.

**Python**

for configuration processing and AI/ML

**FastAPI**

for backend Service

**Web dashboard**

for network administrators

**Netmiko / NAPALM**

for network-device interaction, where applicable

**Initial prototype**

focused on selected vendors and configuration types

**Vendor-agnostic architecture**

for future expansion

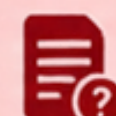
02 CHALLENGES AND RISKS

**Different Vendor Syntax**

The same security requirement may use different commands and structures across vendors.

**Software and Configuration Changes**

Software and firmware updates may change configuration syntax and structures.

**Unknown Configuration Structures**

New vendors or unfamiliar platforms may contain configuration patterns the system does not recognize.

**Growing Network Diversity**

New vendors, operating systems, security standards, and cloud environments can increase compliance complexity.

03 STRATEGIES TO OVERCOME THEM

**Vendor Adapter / Parser**

Handles vendor-specific configurations during the ingestion and normalization process.

**Common Security Model**

Separates vendor-specific syntax from security meaning and applies the same security rules across different vendors

**Reusable Compliance Engine**

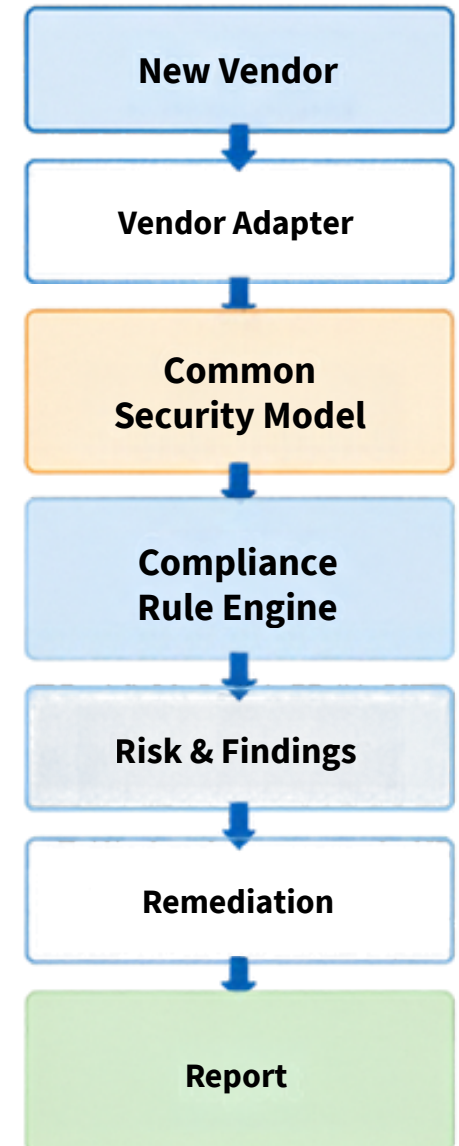
Runs compliance rules against the normalized security model, avoiding a separate auditing system for each vendor.

**Administrator-Guided Learning**

An administrator identifies the meaning of an unfamiliar configuration. The system then maps it to the appropriate security concept.

**Future Adaptation**

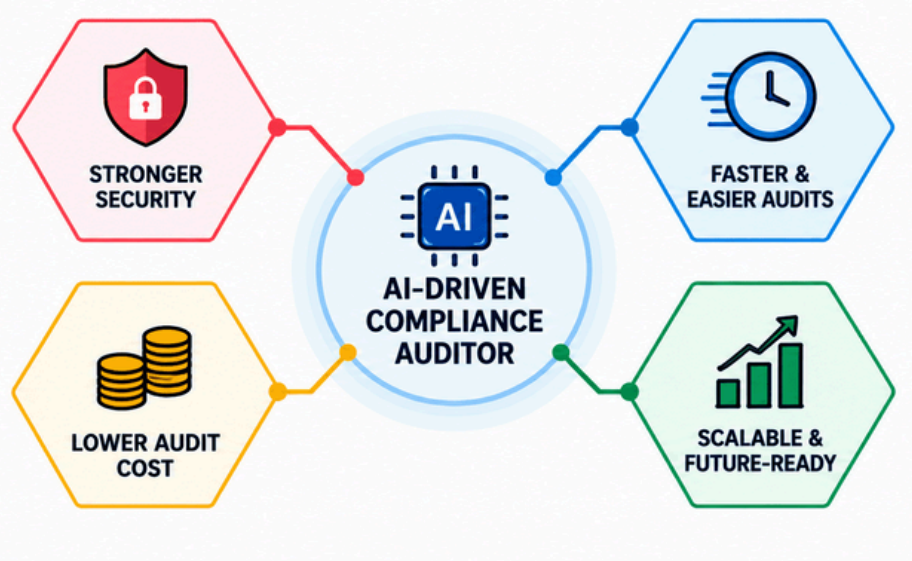
Reuses learned mappings when similar configurations appear in the future.

 SCALABILITY
CONCEPT

KEY TAKEAWAY

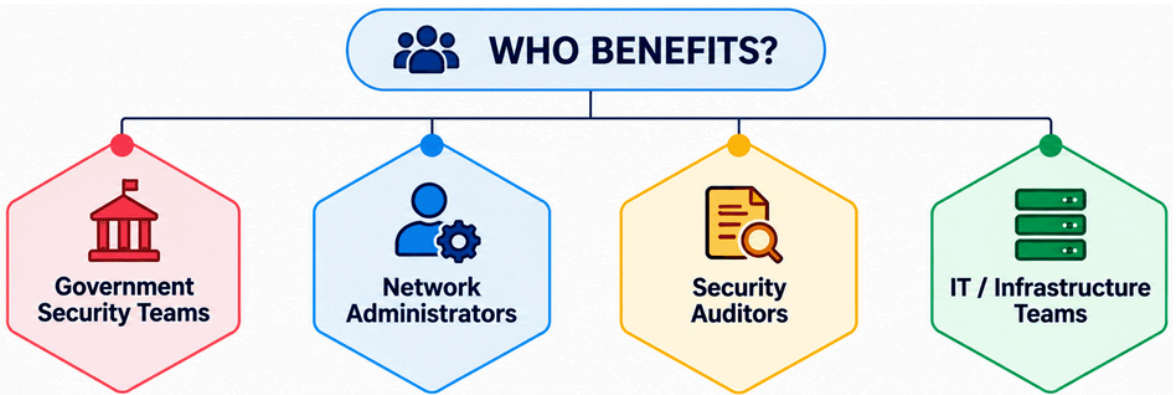
The modular architecture allows new vendors and configuration types to be added without rebuilding the entire compliance system.

Potential Impact on Target Audience



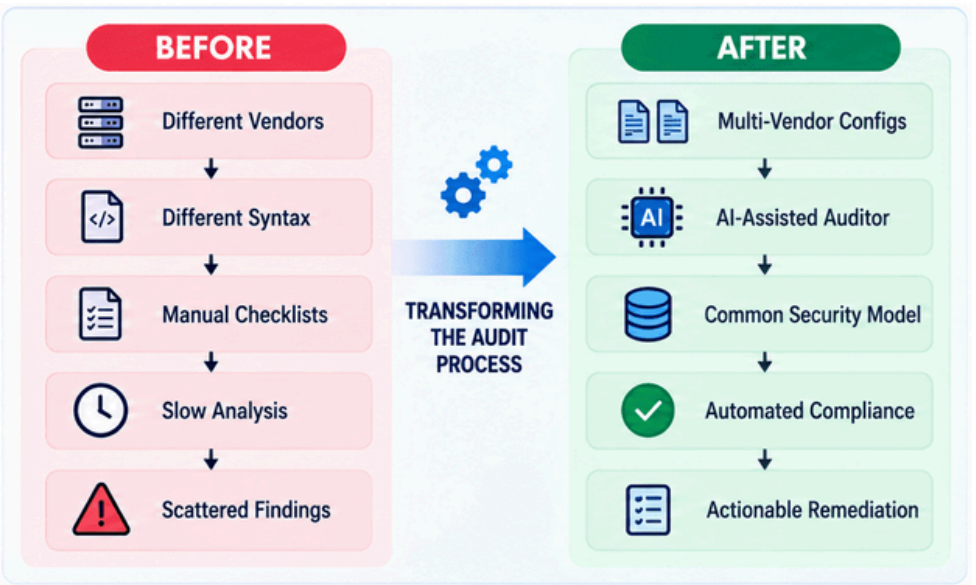
Our solution improves the network security auditing process by making it faster, simpler, and more reliable. Instead of manually checking thousands of configuration lines across different vendors, the AI-driven auditor automatically identifies security gaps, assesses their severity, and provides actionable remediation steps. This helps organizations improve their security posture while reducing the time and effort needed for regular compliance audits. The system is also designed to scale as the network grows and as new vendors or configuration types are added.

Who Gets Benefit?



The main beneficiaries are government security teams, network administrators, security auditors, and IT infrastructure teams. In heterogeneous networks, teams often deal with different vendors, different configuration syntaxes, manual checklists, slow analysis, and scattered findings. Our solution brings these configurations into a single AI-assisted platform, converts them into a common security model, performs automated compliance checks, and presents the results in one centralized view. This shifts the process from simply finding a problem to clearly showing what failed, why it matters, and how it can be fixed.

Before and After



The solution also provides social, economic, and environmental benefits. Socially, it supports better security and stronger compliance across critical network infrastructure. Economically, it reduces manual auditing effort and the need for multiple vendor-specific tools by providing one common platform. From an environmental perspective, digital configuration analysis and reporting reduce paper-based documentation and unnecessary physical audit activities. Overall, our goal is to make network auditing simpler, more consistent, and easier to manage as the network grows.



WIDER BENEFITS



SOCIAL

- Better security posture
- Improved compliance



ECONOMIC

- Lower manual effort
- Reduced tool dependency



ENVIRONMENTAL

- Less paper-based reporting
- Less unnecessary audit travel



One platform



NIST Cybersecurity Framework 2.0

National Institute of Standards and Technology (NIST)

Used as a reference for cybersecurity risk management, security assessment, and communication.

link: <https://www.nist.gov/cyberframework>

CIS Benchmarks

Center for Internet Security (CIS)

Used as a reference for secure configuration and system hardening, including benchmarks for network devices.

link: <https://portal.cisecurity.org/benchmarks>

DISA Security Technical Implementation Guides (STIGs)

Defense Information Systems Agency (DISA)
Referenced as a security hardening and compliance baseline specified in the problem statement.

link: <https://www.cyber.mil/stigs>

ISO/IEC 27001:2022

International Organization for Standardization (ISO)

Used as a reference for information security management and risk management practices.

link: <https://www.iso.org/standard/27001>

Netmiko & NAPALM Documentation

Network Automation Tools

Used as technical references for interacting with and automating multi-vendor network devices. Netmiko supports CLI connections across different vendors, while NAPALM provides a common API for working with different network operating systems.

link: <https://ktbyers.github.io/netmiko/docs/netmiko/index.html>

MITRE ATT&CK

MITRE Corporation

Used as a reference for understanding common attacker tactics and techniques and connecting security findings with real-world attack behavior.

link: <https://attack.mitre.org/>